

Lecture 1: Syllabus, Probability Axioms

Jamie Haddock

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1 Welcome to Math 151

Probability gives us a mathematical language for reasoning about uncertainty – from coin flips and card games to noisy measurements and randomized algorithms. This semester we will build that language up carefully, and test it against **three running examples from randomized numerical linear algebra**: algorithms that use carefully chosen randomness to solve large linear-algebra problems. They will reappear as case studies throughout the course.

1.1 Course logistics

Meeting Tuesdays/Thursdays 1:15-2:30pm, homework weekly via Gradescope (lowest score dropped), two in-class midterms, an in-class final, all details on the syllabus page on Canvas.

2 Probability axioms

Definition: Probability space

A **sample space** Ω is the set of all possible outcomes of an experiment. A **probability law** P assigns a number $P(A)$ to each event $A \subseteq \Omega$.

Theorem: Kolmogorov's Axioms

A probability law P must satisfy:

1. **Non-negativity:** $P(A) \geq 0$ for every event A .
2. **Normalization:** $P(\Omega) = 1$.
3. **(Countable) additivity:** if $A_1, A_2, \dots$ are disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Fact

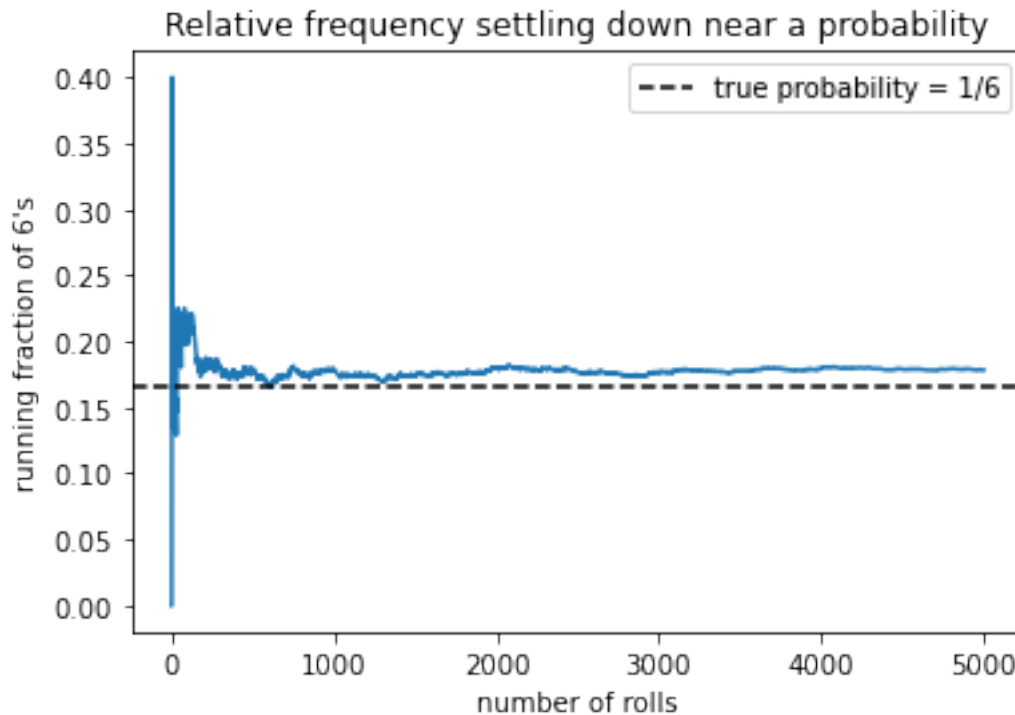
From the axioms alone: $P(\emptyset) = 0$; $P(A^c) = 1 - P(A)$; $A \subseteq B \Rightarrow P(A) \leq P(B)$; and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

2.1 A first look: relative frequency

We haven't defined convergence yet (that's Week 7-8!), but watch what happens to the running fraction of 6's as we roll a die many times.

```
n_rolls = 5000
rolls = rng.integers(1, 7, size=n_rolls)
running_fraction_of_sixes = np.cumsum(rolls == 6) / np.arange(1, n_rolls + 1)

plt.plot(running_fraction_of_sixes)
plt.axhline(1/6, color="black", linestyle="--", label="true probability = 1/6")
plt.xlabel("number of rolls"); plt.ylabel("running fraction of 6's"); plt.legend()
plt.title("Relative frequency settling down near a probability")
plt.show()
```



3 Sneak peek: our three running examples

- **Randomized Kaczmarz** – an iterative solver for $Ax = b$ that samples a random row at each step, according to a carefully chosen distribution.
- **Randomized trace estimation** – estimating $\text{tr}(A)$ for a huge matrix using a handful of random matrix-vector products.
- **Randomized matrix multiplication** – approximating a matrix product by sampling and reweighting a few columns/rows.

No linear algebra background beyond what you already have is needed – we'll introduce exactly what's necessary each time.